Marketing Analytics

Generative AI, Consumer Choice, and Regulation

By Nathan Yang

Session 14

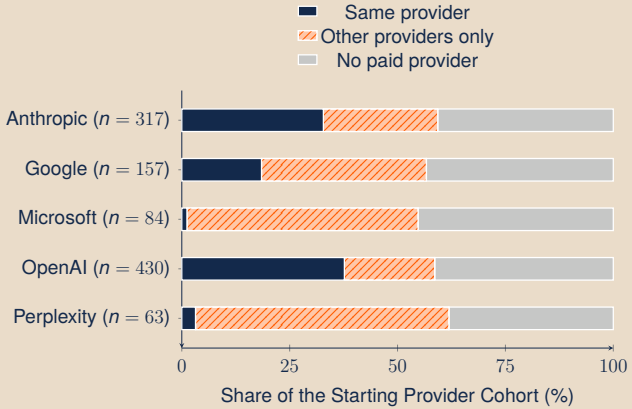
A Subscription-Growth Decision

- A firm wants to increase paid renewals.
- The evidence includes subscription histories, interfaces, and customer outcomes.
- A higher renewal rate can arise through improved value or greater cancellation friction.

Public Attention and Paid Demand

- Pageviews measure attention to selected pages.
- Paid subscriptions measure a specific payment relationship.
- User, provider, and category denominators differ.

A Fixed Starting Cohort

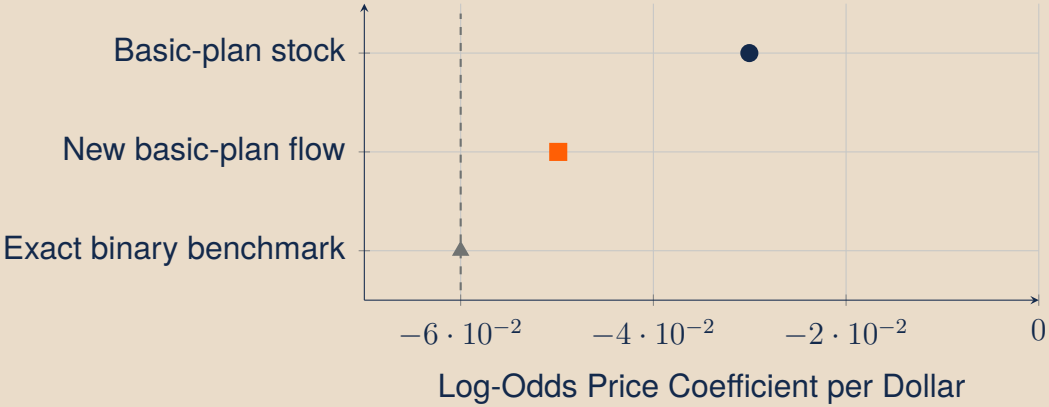


Entirely synthetic subscription panel. Provider labels identify simulated options.

Subscriber Stocks and New Flows

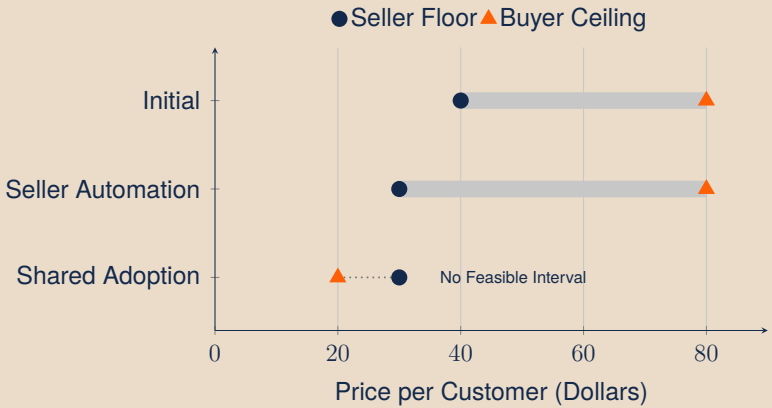
- A subscriber stock includes past acquisition and survival.
- A new-subscription flow can include first purchases and returns.
- A price coefficient depends on the outcome and behavioral model.

A Known Model Is a Useful Verification Case



Synthetic demand exercise with a known binary-model benchmark.

Seller Costs and Customer Alternatives

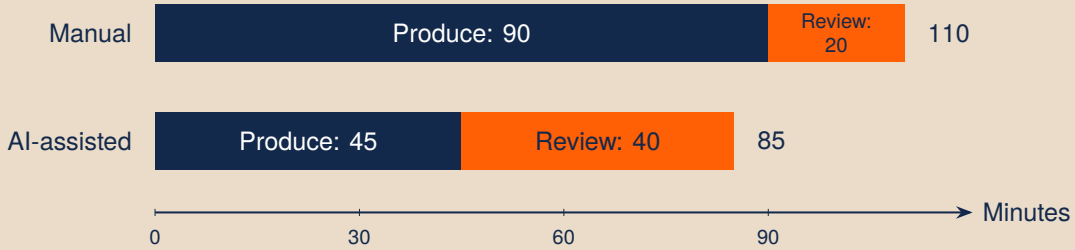


Hypothetical dollars per customer, conditional on 50 sales.

AI Workflow Quality

- Measure the completed analysis and its validity.
- Include generation, review, and repair time.
- Use comparable tasks and independent quality criteria.

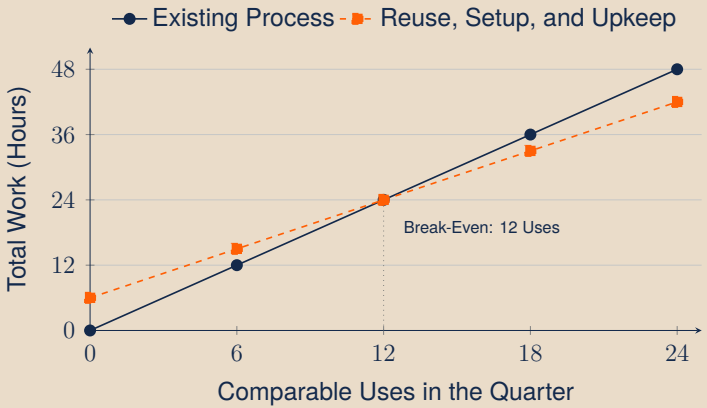
Review Time Is Part of Production



Drafting time falls 50%; total time falls 22.7%.
Task and completed-work quality held fixed.

Synthetic workflows meet the same stated quality requirement.

Reuse Has a Break-Even Workload



Hypothetical equal-quality work: six setup hours, with adaptation included.

Model Updates and Unresolved Tasks

- Identify the changed model, prompt, tool, data, or evaluation rule.
- Compare paired cases and fresh tasks using the same criteria.
- Include regressions, review costs, and customer consequences in the decision.

Residual Risk and Scale

- An expired research category can still allow residual failures.
- Expected failure loss depends on uses, failure probability, and consequences.
- Verify that references exist and support the claims made.

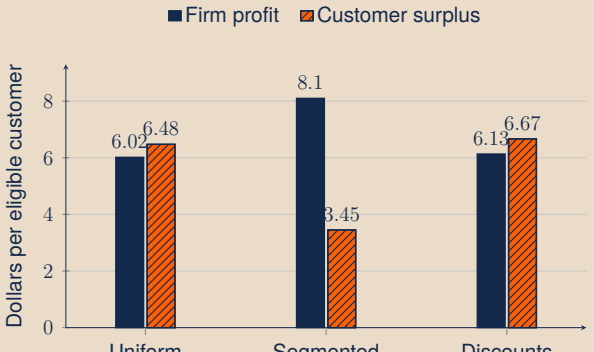
AI Experimentation: Evidence and Safeguards

- Causal Alignment learns from comparisons within past A/B tests.
- Its field result concerns clicks, with human selection and content checks.
- Customer benefit and informed choice require additional evidence.

Personalized Pricing: Benchmark and Incidence

- Dubé and Misra study employers purchasing recruiting services.
- Their model predicts higher profit and lower total customer surplus.
- A majority benefits; the later field comparison uses different price policies.

Majority Gains and Aggregate Customer Losses



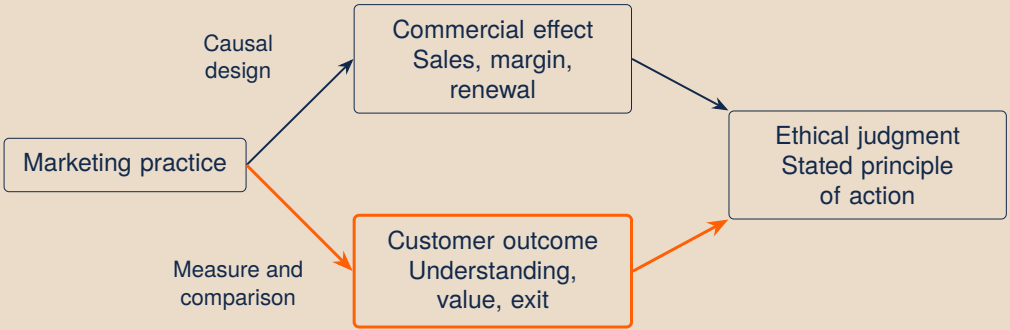
Synthetic population; one purchase opportunity; zero marginal cost.
Segmentation benefits 60% and reduces aggregate customer surplus.

Synthetic population and solved prices; dollars per eligible customer.

Marketing Dialectics: Examine the Premise

- State the claim, its evidence, and the premise connecting them.
- Kant examines reason’s demand for an unconditioned explanation.
- A marketing analogy must preserve the scope of the original argument.

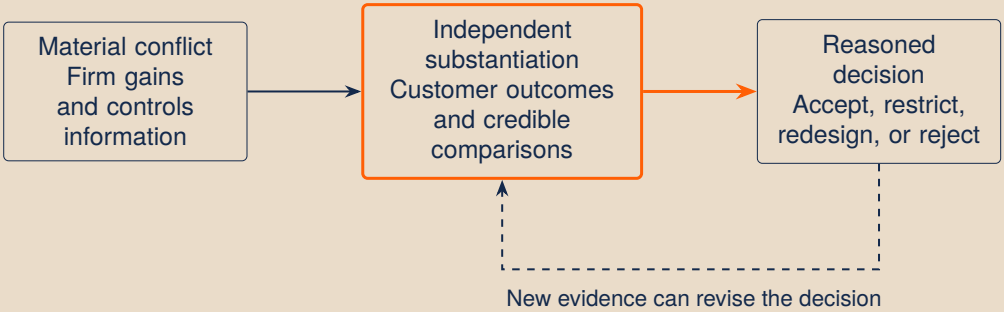
Commercial Results and Customer Claims



A measured sales effect leaves the customer measure and ethical premise substantive.

Causal, measurement, and ethical questions have different premises.

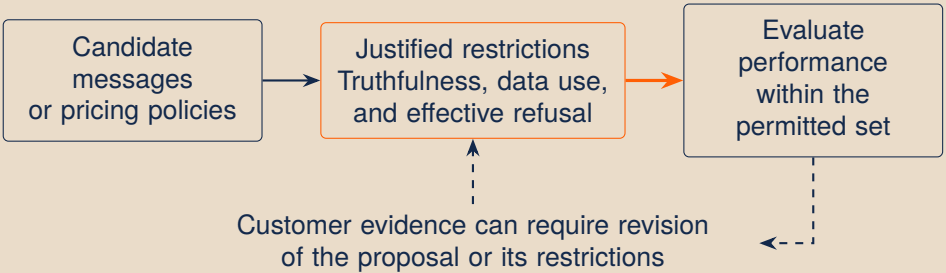
Self-Interest Creates a Burden of Justification



The conflict supplies a reason for scrutiny. The verdict depends on the practice and evidence.

A reason for scrutiny is distinct from evidence of misconduct.

Justify the Practice Before Ranking Its Performance



A higher commercial score supplies a ranking.
An action also requires a justifiable principle and treatment of persons.

A commercial ranking also needs a justified set of permissible actions.

Confucian Perspectives: Profit and Responsibility

- Evaluate how profit is obtained and which relationships sustain it.
- A trusted adviser or customer-care role carries standards of conduct.
- Examine whether the claimed role matches incentives and practice.

Truthful Challenge and Cultivated Judgment

- Harmony can include reasoned disagreement.
- Learning and reflection should discipline an AI-generated answer.
- Good conduct depends on repeated practices as well as stated values.

A Choice Procedure Can Change Renewal

- A direct offer and an offer preceded by questions can produce different choices.
- The customer still makes the final selection.
- An independent value measure determines whether an additional renewal benefits that customer.

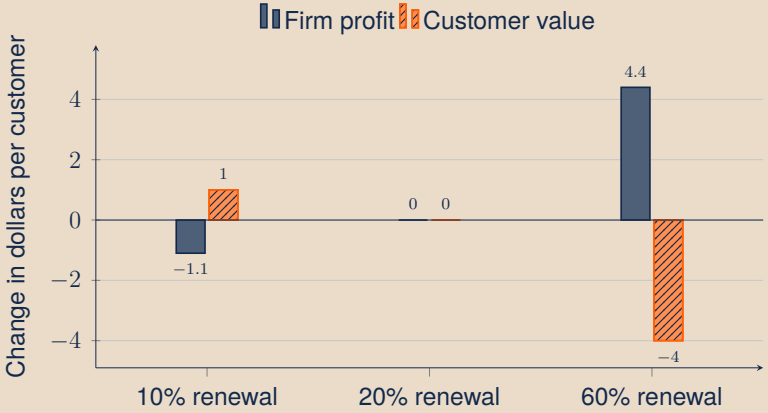
Opposed Interests

Suppose renewal yields firm margin $\nu > 0$ and costs the customer $\sigma > 0$ relative to declining.

$$\Delta\Pi = \nu(s - s_0), \qquad \Delta V = -\sigma(s - s_0)$$

A higher renewal probability raises profit and lowers customer value under these stated stakes.

Profit and Customer Value Can Move Oppositely



Synthetic probabilities and fixed stakes. Direct renewal is 20%.

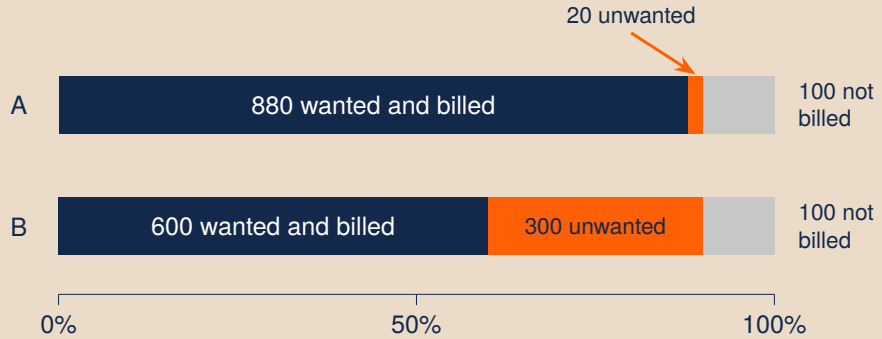
Experimentation Selects the Commercial Winner

- A fixed set of procedures has stable renewal probabilities.
- Independent randomized samples estimate each probability.
- With opposed interests, accurately selecting the largest rate also selects the largest loss in that set.

A Cancellation Evaluation

- Include a direct-cancellation comparison.
- Record whether optional retention content can be refused before exposure.
- Measure completion, elapsed time, subsequent charges, and customer value.

Same Renewal, Different Customer Intent



Synthetic populations: 1,000 customers each, one renewal month.

A Welfare Comparison

Synthetic stakes distinguish profit from customer value.

```
direct, selected = 0.20, 0.60
firm_margin, customer_loss = 11, 10
firm_gain = firm_margin * (selected-direct)
value_change = -customer_loss * (selected-direct)
assert round(firm_gain, 2) == 4.40
assert round(value_change, 2) == -4.00
```

Higher combined surplus can coexist with a customer loss under these assumptions.

Coding Exercise: A Marketing Evaluation

- Reproduce the two populations with identical billed renewal.
- Keep unknown intent separate and compare the relevant denominators.
- State the evidence and ethical premise needed to justify the practice.

Federal Cancellation Policy: Current Status

- July 2025: the Eighth Circuit vacated the 2024 FTC rule.
- February 2026: the FTC restored the earlier prenotification rule.
- March 2026: an advance notice sought input on possible amendments.

Cancellation Rules Differ by Jurisdiction

- California strengthened its Automatic Renewal Law on July 1, 2025.
- New York City’s separate rule took effect October 1, 2026.
- Identify covered transactions, enrollment media, and sector exceptions.

Research Frequency and Policy Importance

- Social influence leads the selected literature clusters.
- Exit friction can affect many customers even with few papers in the sample.
- Choose safeguards using mechanism, exposure, and customer consequences.

Pricing, Reviews, and Delivery

- Price-display rules require attention to the covered industry.
- Reviews presented as customer experiences require truthful provenance.
- Housing-ad delivery can create discrimination concerns beyond targeting choices.

Pricing Language and Regulatory Evidence

- Shelf infrastructure, price rules, and consumer effects are separate objects.
- The FTC’s January 2025 report describes research on pricing intermediaries.
- Specify the practice and applicable authority before making a legal claim.

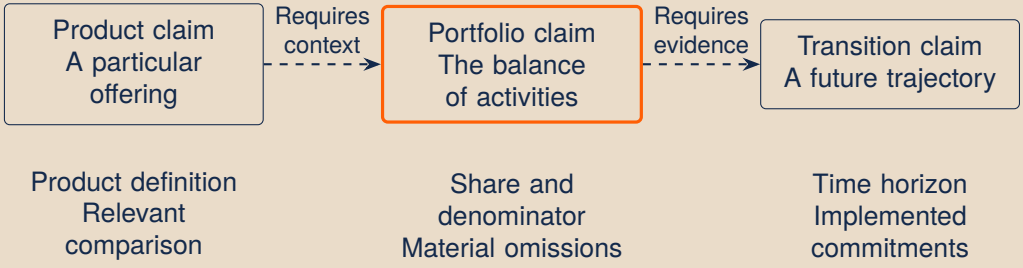
Personalized Pricing: Policy Status

- FTC: August 2026 proposal; October 2026 information-gathering inquiry.
- New York: disclosure law in effect; broader restrictions await enactment.
- Price gouging requires its own covered circumstances and legal elements.

Environmental Claims: Two Different Decisions

- 2023 Shell ads: the ASA upheld the overall-impression complaint.
- 2025 Shell ad: the ASA accepted the contextualized communication.
- Evaluate the particular claim, material context, and applicable standard.

A Product Fact and a Portfolio Impression



A supported narrow statement can invite an unsupported broader inference.

The scope of the conclusion determines the evidence required.

A Decision With Consumer Outcomes

- State the commercial target and customer outcome.
- Identify the procedure the customer can accept, refuse, or revise.
- Connect the proposed action to current rules and supporting evidence.